

Present Value of CF in period T: $PV = \frac{CF_T}{(1+i)^T}$

Present Value of Perpetuity: $PV = \frac{CF}{i}$

Present Value of Growing Perpetuity: $PV = \frac{CF}{i-g}$

Present Value of Annuity: $PV = \frac{CF}{i} \left[1 - \frac{1}{(1+i)^N} \right]$

Present Value of Growing Annuity: $PV = \frac{CF}{i-g} \left[1 - \left(\frac{1+g}{1+i} \right)^N \right]$

Bond Price: $B = \frac{C}{y/m} \left[1 - \frac{1}{\left(1 + \frac{y}{m} \right)^{mT}} \right] + \frac{F}{\left(1 + \frac{y}{m} \right)^{mT}}$, C is periodic coupon, m is payment frequency.

Stock Price: $P_t = \frac{(Div_{t+1} + P_{t+1})}{1+r_e}$

Constant Growth Model: $P_t = \frac{Div_{t+1}}{r_e - g}$, $r_e = \frac{Div_{t+1}}{P_0} + g$

Dividend: $Div_t = Payout Ratio * Earnings_t$

Net working capital: $NWC = Cash + Inventory + Receivables - Payables$

Incremental Earnings = (Incremental Revenue – Incremental Cost – Depreciation) x (1 – Tax Rate)

Free Cash Flow = Incremental Earnings + Depreciation – CapEx – Change in NWC

Scenario Analysis: $E[r] = \sum_s prob(s)r(s)$, $Var(r) = \sum_s prob(s)(r(s) - E[r])^2$, $Std(r) = \sqrt{Var(r)}$

Capital Asset Pricing Model: $r_i = r_f + \beta(r_M - r_f)$, r_f is the risk-free rate, r_M is the expected return of the market.

CAPM Beta of stock i: $\beta = \frac{\sigma_{i,M}}{\sigma_M^2} = \frac{\sigma_i \rho_{i,M}}{\sigma_M}$, where $\sigma_{i,M}$ is the covariance between stock and market, σ_M is the standard deviation of market return, $\rho_{i,M}$ is the correlation between stock i's return and the market return.

Portfolio Returns: $r_p = \sum_i w_i r_i$, where $\sum_i w_i = 1$

Portfolio Betas: $\beta_p = \sum_i w_i \beta_i$, where $\sum_i w_i = 1$

Portfolio Variance: $Var_p = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_{12}$, where $\sigma_{12} = \sigma_1 \sigma_2 \rho_{12}$, $\sigma_p = \sqrt{Var_p}$

Weighted Average Cost of Capital: $WACC = \frac{D}{V} r_{debt} (1 - Tax Rate) + \frac{E}{V} r_e$, where $V = D + E$

$WACC = \frac{D}{V} r_{debt} (1 - Tax Rate) + \frac{E_{preferred}}{V} r_{preferred} + \frac{E_{common}}{V} r_{common}$,
where $V = D + E_{preferred} + E_{common}$

Cost of new equity issuance (assuming constant growth): $r_{external} = \frac{Div_1}{P_0(1-f)} + g$, f is the flotation cost.